

AES encryption

Advanced Encryption Standard

CSE 22 B1 Group 4

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Agenda

1. Introduction

1

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- Advanced Encryption Standard (AES) is the world's most widely used encryption algorithm
- Adopted by US government in 2001 to protect sensitive information
- Replaced the aging Data Encryption Standard (DES)
- Used in everything from secure web browsing (HTTPS) to encrypted messaging and data storage
- Foundation of modern digital security

1

Introduction

Why AES Matters:

AES secures billions of online transactions, communications, and data storage operations every day. By doing so, it helps create a safer digital world where privacy and trust are possible.

2. Understanding Encryption

2

What is Encryption?

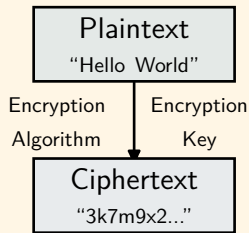
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- Hides information as secret code
- Only the right key can decrypt



Encryption process

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Why Encryption Matters

- **Confidentiality:** Stolen data stays unreadable without the key.
- **Privacy:** Protects messages, personal data, and transactions from eavesdropping.
- **Compliance:** Needed for standards/laws (e.g., GDPR, HIPAA, PCI-DSS).

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- **Threat defense:** Helps against breaches, MITM, and ransomware fallout.
- **Trust & integrity:** Reduces tampering risk and increases confidence.

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Application of Encryption

Secure Communications

HTTPS, encrypted messaging (Signal/WhatsApp)

Data Storage

Disk + cloud encryption

Financial Transactions

Online banking, card payments

Authentication

Password hashing, secure tokens

Virtual Private Networks

Encrypted traffic for privacy/security

Digital Rights

Prevents unauthorized copying/use

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3. Characteristics of Good Encryption

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Need for understanding good encryption criteria

Before AES, it helps to know what makes an encryption standard *good*. These criteria explain why AES is designed the way it is.

3 Criteria for Good Encryption

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specified design

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(Shannon's Principles)**

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**Resistance to known
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**Efficiency in software
and hardware**

3.1 Public, Well-Specified Design

A good standard is openly documented (algorithm, parameters, test vectors) so experts can review it, implement it consistently, and find flaws early.

Key Point Transparency enables peer review and builds trust.

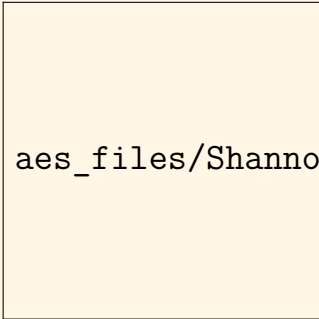
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Strong Design Principles

Shannon's Principles

A robust encryption standard follows Shannon's core principles:

- Confusion
- Diffusion
- Adequate Key Size



aes_files/Shannon.jpg

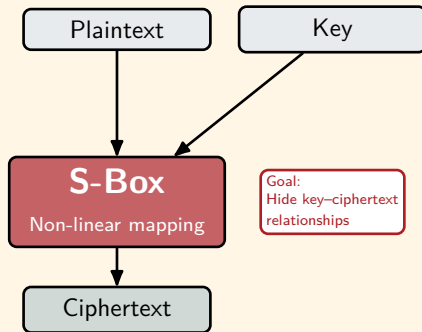
Claude E. Shannon

3.2

Shannon's Principle: Confusion

Make the plaintext–ciphertext–key relationship hard to predict.

Example Non-linear substitutions (S-boxes) make key recovery from patterns difficult.



3.2 Shannon's Principle: Diffusion

Small input changes should spread widely in the output.

Avalanche Effect Flip 1 plaintext bit → about half the ciphertext bits change.

Plaintext:

1 0 1 0 1 1 0 1

Diffusion Layer

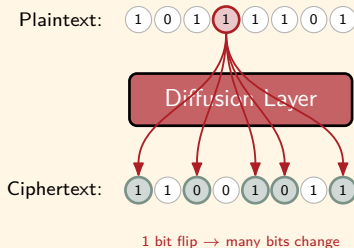
Ciphertext:

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3.2 Shannon's Principle: Key Size

Keys must be large enough to make brute-force search impractical.

Modern Standards

- 128-bit: 2^{128} keys
- 256-bit: 2^{256} keys
- Exhaustive search is infeasible today

3.3 Large Key Space & Key Handling

Beyond "big keys," a good standard defines safe key formats/schedules and practical guidance (generate, store, rotate) to avoid weak or predictable keys.

Warning Poor key management breaks even strong algorithms.

3.4 Resistance to Known Attack Classes

A strong standard is tested against known cryptanalysis and keeps a practical security margin.

- Differential cryptanalysis
- Linear cryptanalysis
- Side-channel attacks
- Brute-force attacks

3.5 Efficiency in Software & Hardware

If encryption is too slow, people disable or misuse it; a good standard is fast on common devices.

Practical Requirements

- Real-time throughput (e.g., streaming)
- Low power on mobile
- Hardware acceleration support

4. Rijndael Algorithm

4.1 What is Rijndael?

- **Rijndael** is the algorithm underlying AES (Advanced Encryption Standard).
- Designed by Joan Daemen and Vincent Rijmen.
- A symmetric block cipher with variable block and key sizes.
- Selected as the AES standard by NIST in 2001.

Key Characteristics Block size: 128 bits | Key size: 128, 192, or 256 bits

4.1 AES Core Operations

- **Key Expansion**
- SubBytes
- ShiftRows
- MixColumns
- AddRoundKey

4.2 Key Expansion Process

- Expands cipher key into round keys.
- Number of round keys depends on key size:
 - 128-bit key: 11 round keys
 - 192-bit key: 13 round keys
 - 256-bit key: 15 round keys
- Uses **RotWord**, **SubWord**, and **Rcon** functions.

Rcon Values Round-dependent constants used in key schedule to add asymmetry.

4.2 Simplified Key Expansion Algorithm

```
def key_expansion(key, rounds):  
    w = []  
    # Copy original key  
    for i in range(4):  
        w.append(key[i])  
  
    # Generate round keys  
    for i in range(4, 4*(rounds+1)):  
        temp = w[i-1]  
        if i % 4 == 0:  
            temp = SubWord(RotWord(temp))  
            temp ^= Rcon[i//4]  
        w.append(w[i-4] ^ temp)  
    return w
```


4.2 Key Expansion: Detailed Process

The Objective

- Transform the *Cipher Key* into a *Key Schedule*.
- AES-128 requires **176 bytes** (11 keys of 16 bytes each).

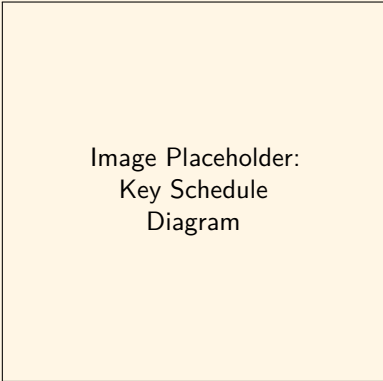


Image Placeholder:
Key Schedule
Diagram

Recursive Key Generation

4.2 Key Expansion: Detailed Process

The Core Transformation (g function)

1. **RotWord**: One-byte left circular shift.
2. **SubWord**: Non-linear byte substitution using S-Box.
3. **Rcon**: XOR with a round-dependent constant to eliminate patterns.

Diffusion A change in one bit of the original key affects multiple bits in all subsequent round keys.

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$\underbrace{\hspace{1.5cm}}_{W_0}$ $\underbrace{\hspace{1.5cm}}_{W_1}$

0x A7

0x 3B

- Initial state in binary (16 bits total).
- Grouped into two 8-bit **words** (W_0 and W_1).
- Converted to Hexadecimal for the Expansion algorithm.

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4.2 Key Expansion: Detailed Process

Initial Key: 0xA73B Split into words: $W_0 = A7$, $W_1 = 3B$

Step 1: Generate W_2

- **RotWord(W_1):** $3B \rightarrow B3$
- **SubWord:** $B3 \rightarrow 17$ (via S-Box)
- **Rcon:** $17 \oplus 80 = 97$
- **Result:** $W_0 \oplus 97 = 30$

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Step 2: Generate W_3

- **Logic:** $W_2 \oplus W_1$
- **Calculation:** $30 \oplus 3B$
- **Result:** $0B$

Round Key 1 $W_2W_3 = 0x300B$

This process repeats using Round Key 1 to generate Round Key 2, and so on, until all round keys are derived.

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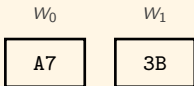
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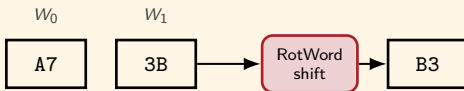
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Key Expansion: Step-by-Step Visualization



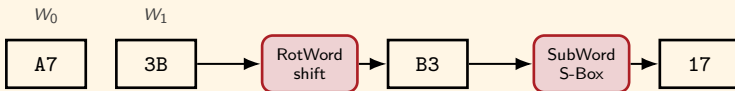
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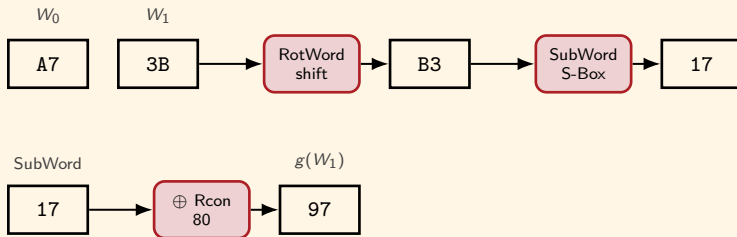
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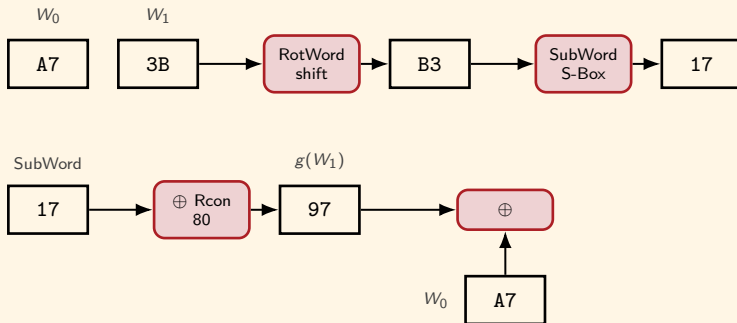
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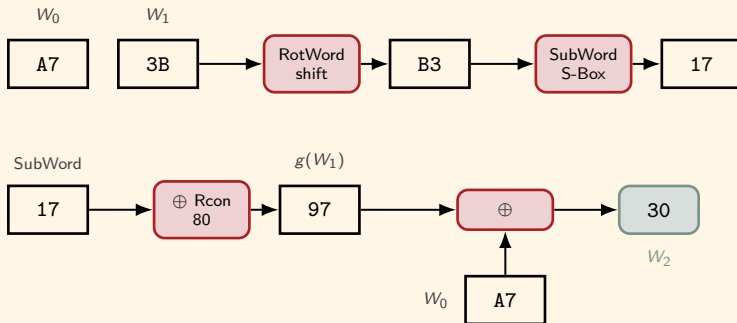
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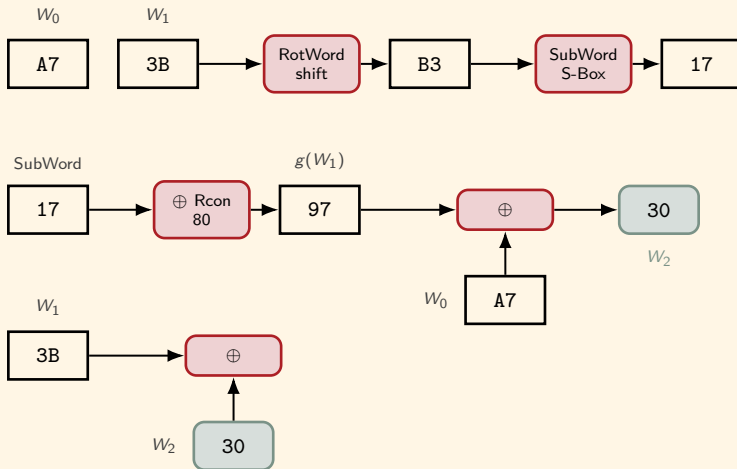
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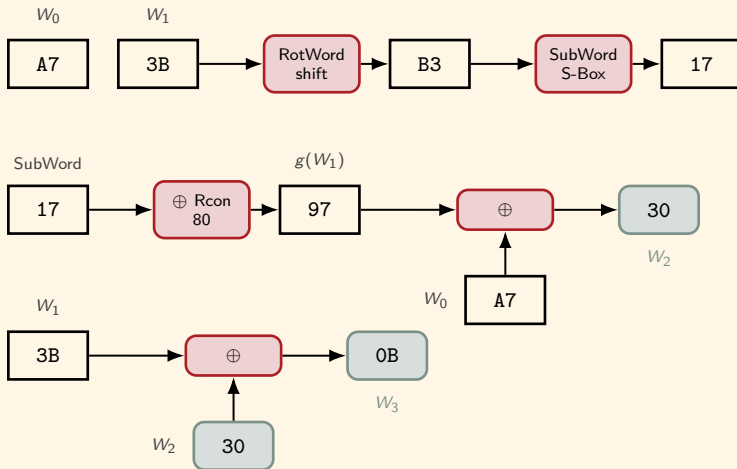
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5. Analysis of AES

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AES vs Good Encryption

AES perfectly aligns with the criteria for good encryption:

Public Design: Open NIST competition ensured review.

Shannon's Principles: Exceptional confusion and diffusion.

Key Space: Massive keys ensure brute-force resistance.

Attack Resistance: Highly resilient against mathematical shortcuts.

Efficiency: High speed across all types of devices.

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The Crucible: Instead of a secret design, NIST launched a 5-year open global competition with 15 candidates.

The Victory: In 2000, *Rijndael* won, defeating submissions from industry giants like IBM and RSA.

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Global Trust Through Transparency

Proven Standard: Published openly as FIPS 197 and later approved by the NSA for 'Top Secret' data, proving its mathematical resilience.

The Takeaway: Kerckhoffs's Principle A cryptosystem must be secure even if everything about it (except the key) is public knowledge. AES earns trust not by hiding its mechanics, but by mathematically surviving decades of peer review.

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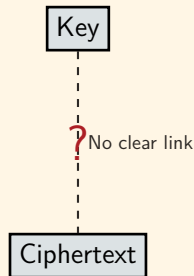
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AES achieves confusion primarily through the SubBytes operation.

It utilizes a highly non-linear S-Box based on the multiplicative inverse in the Galois Field $GF(2^8)$.

The Result The relationship between the ciphertext and the symmetric key is made incredibly complex, thwarting mathematical pattern recognition.

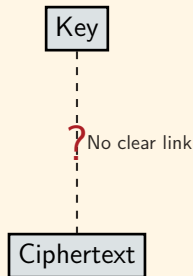


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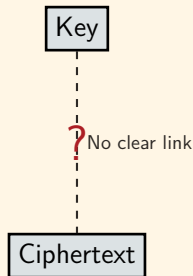


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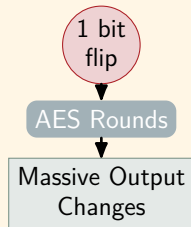


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The Avalanche Effect Statistically, flipping just a single bit in the 128-bit plaintext block alters approximately 50% of the resulting ciphertext bits.

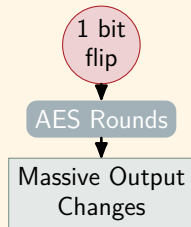


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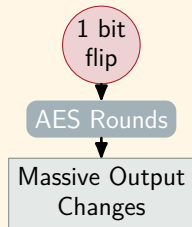


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A Massive Key Space

AES mandates key lengths of 128, 192, or 256 bits, exponentially increasing the computational complexity of exhaustive search.

A standard 128-bit key yields 2^{128} possible combinations (over 340 undecillion).

A robust Key Expansion algorithm guarantees that every derived round key is cryptographically secure and unique.

128-bit Key

192-bit Key



256-bit Key

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5 Resistance to Cryptanalysis

AES was explicitly designed to mathematically neutralize established cryptanalytic techniques.

The algebraic complexity of the S-Box provides proven immunity against Differential and Linear Cryptanalysis.

The rigorous key schedule mitigates vulnerabilities against Related-Key Attacks.



AES Math Wall

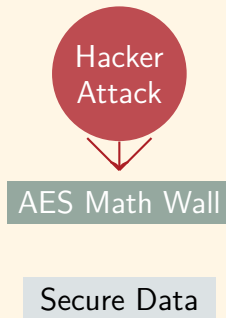
Secure Data

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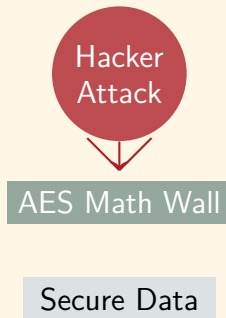


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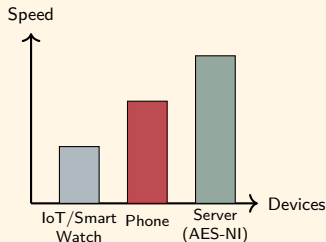
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Efficiency in Software & Hardware

AES avoids computationally heavy math, relying exclusively on fast operations like XOR and bit shifts.

It features widespread hardware acceleration; modern CPUs execute it directly at the silicon level via instruction sets like AES-NI.

The Result: Highly scalable performance, running seamlessly on everything from constrained IoT devices to enterprise servers.



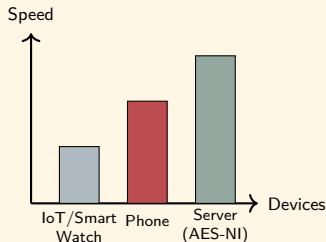
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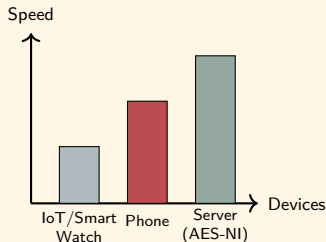
5

Efficiency in Software & Hardware

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6. Conclusion

6

The Global Standard

Ubiquity: AES is the trusted standard securing daily digital life, from local Wi-Fi networks to international banking.

Cryptographic Elegance: It flawlessly implements Shannon's principles of Confusion and Diffusion through simple but highly effective byte math.

Future-Proof: The AES-256 variant provides a robust security margin, widely considered safe even against future quantum computing threats.

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Final Thoughts

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It is the fundamental cornerstone of trust in our modern digital society.

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